

SMARTRESUME AI
AI-Powered Resume Builder
AI Prompt Engineering Workshop – Final Project

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Department: MCA – Master of Computer Applications

Project Title: Smart Resume AI

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Project Description –

Smart Resume AI is an Artificial Intelligence-powered web application developed using Python and Streamlit. The application helps users generate professional, ATS-friendly resumes by collecting personal, educational, and professional information and using an AI language model to create a well-structured resume.

The project demonstrates the practical use of Prompt Engineering by converting user inputs into optimized prompts that enable the AI model to generate high-quality resume content. The generated resume can be previewed within the application and exported as PDF or Microsoft Word (DOCX), making it suitable for job applications and internships.

AI Tool(s) Used –

Tool	Purpose
ChatGPT (OpenAI API)	Generate professional ATS-friendly resume content
Python	Backend programming
Streamlit	Web application interface
Visual Studio Code	Development environment
python-docx	Export resume as Microsoft Word
FPDF2	Export resume as PDF

Prompts Used –

System Prompt –

You are an expert Resume Writer and Career Coach.

Your responsibility is to create professional ATS-friendly resumes.

Rules:

- Use formal English.
- Keep the resume concise.
- Highlight strengths.
- Use bullet points wherever appropriate.
- Do not invent information.
- Improve grammar.
- Make the resume suitable for recruiters.
- Return only the resume.

User Prompt –

Generate a professional ATS-friendly resume using the following information.

Name: John Doe

Email: johndoe@example.com

Phone: +91 9876543210

Education: Bachelor of Engineering in Computer Science

Skills: Python, Java, SQL, HTML, CSS, JavaScript

Experience: Software Development Intern

Projects: Smart Resume AI & Library Management System

Certifications: AI Prompt Engineering & Python Programming

Languages: English, Hindi, Kannada

Career Objective: Seeking an entry-level Software Engineering position where I can apply my programming skills, prob

Results –

Application Features –

The SmartResume AI application provides the following features:

- User-friendly web interface
- Personal information collection
- Education details
- Technical skills
- Work experience
- Academic and personal projects
- Certifications
- Languages
- AI-powered resume generation
- Resume preview
- Export as PDF
- Export as Microsoft Word (DOCX)

Sample AI-Generated Resume –

JOHN DOE

Email: johndoe@example.com

Phone: +91 9876543210

Career Objective –

Motivated Computer Science graduate with strong programming knowledge and practical experience in Python, Java, SQL, and Artificial Intelligence. Passionate about software development and eager to contribute to innovative projects while continuously learning new technologies.

Education –

Bachelor of Engineering in Computer Science

XYZ Institute of Technology

CGPA: 8.75

Graduation Year: 2026

Technical Skills –

- Python
- Java
- C
- C++
- HTML
- CSS

Work Experience –

Software Development Intern

ABC Technologies

- Developed Python automation scripts.
- Fixed application bugs.

- Improved software performance.
- Collaborated using Git and GitHub.

Projects –

SmartResume AI

- Developed an AI-powered resume builder using Python and Streamlit.
- Integrated AI for resume generation.
- Added PDF and DOCX export functionality.

Library Management System –

- Developed using Python and MySQL.
- Implemented authentication and database management.
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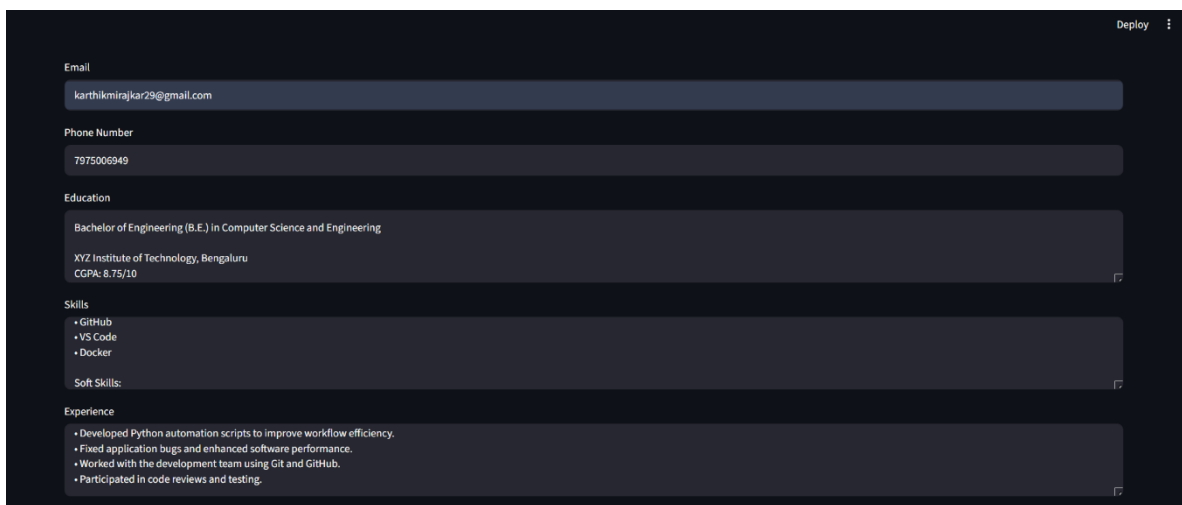
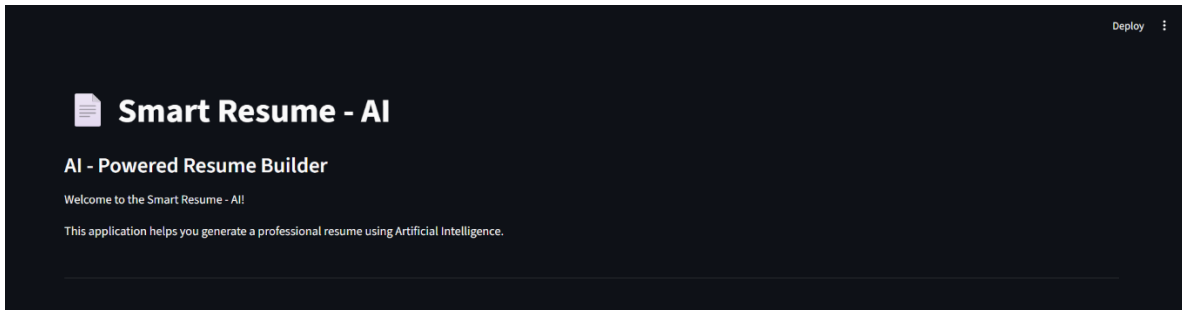
Certifications –

- AI Prompt Engineering Workshop
- Python Programming
- SQL Fundamentals
- Git & GitHub Essentials

Languages –

- English
- Hindi
- Kannada

Screenshots —



Deploy

Preview

Karthik Mirajkar

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Education

Bachelor of Engineering (B.E.) in Computer Science and Engineering

XYZ Institute of Technology, Bengaluru CGPA: 8.75/10 Graduation Year: 2026

Relevant Coursework: • Data Structures & Algorithms • Database Management Systems • Operating Systems • Computer Networks • Artificial Intelligence • Machine Learning

Deploy

Skills

Programming Languages: • Python • Java • C • C++

Web Technologies: • HTML • CSS • JavaScript

Databases: • MySQL • MongoDB

Tools: • Git • GitHub • VS Code • Docker

Soft Skills: • Problem Solving • Teamwork • Communication • Time Management

Experience

Software Development Intern ABC Technologies January 2025 – June 2025

• Developed Python automation scripts to improve workflow efficiency. • Fixed application bugs and enhanced software performance. • Worked with the development team using Git and GitHub. • Participated in code reviews and testing.

Projects

SmartResume AI • Developed an AI-powered resume builder using Python, Streamlit, and OpenAI API. • Integrated AI to generate professional ATS-friendly resumes. • Added PDF and DOCX export functionality.

Library Management System • Built using Python and MySQL. • Implemented book issue/return and user authentication.

Student Attendance System • Developed using Python and SQLite. • Automated attendance recording and report generation.

Deploy

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Certificates

• AI Prompt Engineering Workshop • Python Programming Certification • SQL Fundamentals • Data Structures and Algorithms • Git & GitHub Essentials

Learning Reflection –

During this project, I gained practical knowledge of Prompt Engineering and learned how carefully designed prompts improve the quality of AI-generated responses. I understood how Large Language Models can be integrated into Python applications using APIs and how Streamlit can be used to build interactive web applications. I also learned to export generated content into PDF and Word formats, making the application more useful in real-world scenarios. Among the AI tools used, ChatGPT was the most effective because it generated professional, well-structured resume content with minimal prompt refinement. Overall, this project enhanced my understanding of AI application development and strengthened my skills in Python, prompt engineering, and software design.